COORDINATING CROWDS AT SHARED FACILITIES: CONGESTION GAMES, INFORMATION DESIGN AND CORRELATED EQUILIBRIUM *

FINAL REPORT | FOUNDATIONS OF AI – MULTI-AGENT SYSTEMS | IIT GANDHINAGAR

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ABSTRACT

Many people share one bottleneck and each chooses *when* to use it: cars at a bridge, backups on a network link, diners at a food court, students at a hostel mess. We model this as an atomic congestion game over time slots with a fluid queue, and treat the crowd display as an *information designer*. We (i) compute Nash equilibria, the social optimum and the price of anarchy (PoA) for five domains and 24 synthetic mess instances (4 messes, 3000 students, 3 meals); (ii) simulate day-to-day play under nine information policies and three behaviour regimes; and (iii) study private recommendations as *correlated equilibria* (CE). Selfish timing costs 3–8× the optimum. An exact live counter that everyone reacts to raises the cost about 5× (the El Farol effect). Recommendations drawn from a Nash schedule are obeyed, stable and remove herding once 50–70% of students use them, but we prove and verify by linear programming that advice alone cannot beat Nash in a large crowd. The CE obedience constraints, however, give the smallest reward that makes the optimum self-enforcing: rewards worth about 2.7 minutes for 13% of students cut the average cost from 2.24 to 0.36 minutes.

Keywords Congestion games · Correlated equilibrium · Information design · Price of anarchy · No-regret learning

1 Introduction

Congestion at a shared facility is usually a *coordination* problem, not a capacity problem. On a popular night the IITGN mess is packed at 20:00–20:30 and almost empty after 21:00, yet the counter could serve everyone over the two hours. The same structure appears on roads (everyone wants to cross a bridge just before 9:00), in networks (laptops all back up at 22:00), in restaurants and in gyms. Every person who joins a queue delays everyone behind them but feels only their own wait.

A live crowd counter, easy to build from the QR scans the mess already records, looks like the obvious fix. But in the El Farol bar problem [1] people who react to the same forecast move together, and a counter could make the rush jump between slots. Our central question is: *what should a display show, how fresh and how accurate should it be, and to whom, so that people's own choices lead to short waits?* Following our advisor's suggestion we treat congestion in general, use the correlated equilibrium [2] as the main solution concept for "telling different people different things", and use the mess as a detailed case study with synthetic data.

Contributions. (1) One slot-congestion model and solver suite (Nash, Wardrop, optimum, PoA) covering roads, networks, restaurants, gyms and the mess. (2) A day-to-day simulator comparing nine information policies under three behaviour models and three learning rules. (3) A population-scale treatment of correlated recommendations: a fair shuffled recommender, a proof that advice cannot beat Nash in large monotone games, a best-CE linear program, and the *price of obedience*. (4) A synthetic semester of QR-scan logs with calibration and nowcasting. Every number below is produced by `python experiments.py`.

* *Code and data generator*: https://github.com/YashG1836/crowd-congestion; *live demo*: https://crowd-congestion.streamlit.app/. Every figure and number in this report is produced by `python experiments.py`.

2 Preliminaries

Definition 1 (Game, NE)**.** *Players i choose actions s_i; $c_i(s)$ is i's cost. A profile is a* Nash equilibrium *(NE) if no player can lower their cost by changing only their own action [3]. Congestion games have pure NE because they admit a potential [4].*

Definition 2 (CE and CCE [2])**.** *A mediator draws $s \sim p$ and privately tells each player s_i. p is a* correlated equilibrium *if for all i and actions a, a': $\sum_{s:s_i=a} p(s)\,[c_i(s) - c_i(a', s_{-i})] \le 0$. It is a* coarse *CE if for all i, a': $\sum_s p(s)\,[c_i(s) - c_i(a', s_{-i})] \le 0$. NE⊂CE⊂CCE, and both sets are polytopes, so the best one is a linear program.*

Definition 3 (PoA)**.** *With social cost $SC = \sum_i c_i$,* PoA $= SC(\text{worst NE})/SC(\text{optimum})$*; the price of stability uses the best NE [5, 6].*

Definition 4 (Incomplete information, learning)**.** *In a Bayesian game players do not know a state ω and best-respond in expectation (Bayesian NE) [7]. In logit response a slot is chosen with probability $\propto e^{-\lambda\, cost}$ [8]. Hedge [9] (full feedback) and EXP3 [10] (bandit feedback) have no external regret, so joint play approaches the CCE set; regret matching [11] has no swap regret and approaches the CE set.*

Three small games (Table 1, Fig. 1). *Pigou's network:* with a peak road costing x^p and an alternative costing 1, $\mathrm{PoA}(p) = [1 - p(p+1)^{-(p+1)/p}]^{-1}$ (4/3 at p=1, 2.15 at p=4). *Two-student dinner game:* Peak costs the number of students there, Late costs 1.5; the shared counter makes both students alternate between (Peak, Peak) and (Late, Late); a private fair coin is a CE that reaches the optimum. *Intersection (Chicken):* utilities 6/6, 2/7, 7/2, 0/0; the best CE ($\frac{1}{2}$ Stop–Stop, $\frac{1}{4}$ each Stop–Go, Go–Stop) gives **5.25** to each driver, more than any mixture of Nash equilibria (4.67).

Table 1: Two-student dinner game: what each kind of information achieves (computed by `theory.py`).

Policy	Total cost	Ratio to optimum
Shared live counter (both react to yesterday)	3.5	1.40
No information (symmetric mixed NE, Peak w.p. 1/2)	3.0	1.20
Private fair coin = correlated equilibrium	**2.5**	**1.00**

3 Problem formulation

Players and slots. n people, T slots of Δ minutes (mess: T=12, Δ=10). Person i has a private preferred slot τ_i and flexibility β_i (minutes of waiting they would accept to avoid moving one slot). People with equal (τ, β) form a *type* k; the state is $X[k,t]$, the number of type-k people in slot t, with loads $A_t = \sum_k X[k,t]$. The cost of a type-k person in slot t is

$$c_k(t; A) = g_t(A) + \beta_k\,|t - \tau_k|, \qquad SC(X) = \sum_t A_t\, g_t(A) + \sum_{k,t} X[k,t]\,\beta_k |t - \tau_k|. \tag{1}$$

Fluid queue. One counter serves μ people per minute, arrivals are spread evenly within a slot, service is first come first served, and leftover queue carries over. With backlog q_0 and net rate $r = (A_t - \mu\Delta)/\Delta$, the queue after u minutes is $\max(0, q_0 + ru)$ and an arrival then waits that divided by μ. Averaging over the slot,

$$W_t = \begin{cases} \left(q_0 + \frac{A_t - \mu\Delta}{2}\right)/\mu & r \ge 0 \\ \left(q_0 s + \frac{r s^2}{2}\right)/(\Delta\mu),\ s = \min(\Delta, q_0/(-r)) & r < 0 \end{cases}, \qquad q_0^{\text{next}} = \max(0, q_0 + A_t - \mu\Delta). \tag{2}$$

Total waiting $\sum_t A_t W_t$ equals the area under the queue (Little's law). This is Vickrey's bottleneck model [12] for roads and a deterministic link buffer for networks. We use $g_t = W_t$, optionally plus a penalty for missing a scarce special item. The game is *atomic*: a deviation from t to t' changes loads to $A - e_t + e_{t'}$.

Information designer. The display chooses a signal policy π (what it shows, how delayed or noisy, and to whom) to minimise the long-run average social cost; we report $SC(\pi)/SC(\text{opt})$ because some policies never settle. **Assumptions** (unchanged from our brief): each student eats once in their own mess and chooses only the time (A1); a scan marks the end of waiting (A2); preferences are private and stable (A3); the policy is public, and students see the signal but not others' choices (A4).

4 Methodology

4.1 Equilibrium, optimum and price of anarchy

Algorithm 1 Best-response dynamics returning an (ε-)Nash equilibrium (`game.nash_equilibrium`)

start from X_0 (everyone at τ; the rounded Wardrop solution; random; the optimum)
repeat
  $G[k,t,t'] \leftarrow c_k(t;A) - c_k(t';A - e_t + e_{t'})$ for all $X[k,t] > 0$ ▷ exact one-person gains
  if $\max G \leq 0$ **return** X (exact NE)
  record X if $\max G$ is the smallest so far; move one person along $\arg\max G$
until a state repeats (cycle) or the iteration cap
return the recorded state, with $\varepsilon =$ its largest remaining gain

Queue carry-over destroys the potential function, so best response can cycle. We therefore return an ε-NE and report ε (worst case over all 24 mess instances: **14 seconds**). The large-population (Wardrop) equilibrium is found by the method of successive averages, $x \leftarrow x + (\mathrm{BR}(x) - x)/(r+1)$ (fictitious play), and used as a starting point. The **social optimum** solves an LP relaxation on a one-minute grid (variables $x[k,t]$ and queue lengths q_m, constraints $q_m \geq q_{m-1} + a_m - \mu$, objective $\sum_m (q_{m-1} + q_m)/2$ plus schedule cost), rounds it while keeping type counts, and polishes it with exact local search (one-person moves and two-person swaps). Local search is also started from the NE, so the reported optimum is never worse than the NE. PoA compares only equilibria whose ε is within 6 s of the best one.

4.2 Correlated recommendations at population scale

The app picks a schedule X and, *within each type*, shuffles who is told which slot. A type-k person is told t with probability $X[k,t]/n_k$ and knows nothing else. Writing C_j, D_j for current and deviation costs under schedule X_j, a randomisation p over schedules is a CE iff $\sum_j p_j X_j[k,t]\,(C_j[k,t] - D_j[k,t,t']) \leq 0$ for all k, t, t'.

Proposition 1 (Shuffled Nash is a CE)**.** *If X is an NE, the shuffled recommendation is a CE.* Proof. *Every realised assignment has type-level schedule X; a person told t knows the loads are $A(X)$, and t is a best response for their type.* □

Proposition 2 (Advice cannot beat Nash in a large crowd)**.** *In the non-atomic limit with monotone congestion costs, every CCE (hence every CE) has the social cost of the Wardrop equilibrium x^*.* Proof sketch. *Weighting the CCE inequalities by x^* gives $\sum_\omega p_\omega \langle x^\omega - x^*, c(x^\omega)\rangle \leq 0$. Each term is $\langle x^\omega - x^*, c(x^\omega) - c(x^*)\rangle \geq 0$ (monotonicity) plus $\langle x^\omega - x^*, c(x^*)\rangle \geq 0$ (variational inequality), so both vanish: the loads, and hence the cost, equal those of x^*.* □

Proposition 3 (Price of obedience)**.** *For the optimum X°, let $G[k,t] = \max_{t'}(C[k,t] - D[k,t,t'])^+$. A reward $b[k,t] = G[k,t]$ for arriving in the recommended slot (e.g. a priority token) makes the shuffled optimum a CE of the rewarded game, at total transfer $\sum_{k,t} X^\circ[k,t]\,G[k,t]$.*

Our queue has carry-over, so Proposition 2 is not exact. We therefore also solve a **best-CE LP** over 26 candidate schedules (Nash equilibria, the optimum, and 20 schedules between them), with an ε slack in each obedience constraint, plus its CCE version.

Incomplete information. If some nights are slow (state ω, prior P) and nobody announces it, strategies cannot depend on ω, and the Bayesian NE is the NE of the game with $g = \sum_\omega P(\omega) g_\omega$. With an announcement (or a mediator who knows ω) people play the NE of the true night.

4.3 Day-to-day simulator, policies and behaviour

The same people choose every day (one day = one meal; 150 days, 50 burn-in, 30 seeds, 95% CIs). The policies are π_0 no information (own memory of visited slots); π_1 exact crowd cost of the last meal; π_2 the same, 3 meals old; π_3 plus private Gaussian noise (sd 3 min); π_4 3 levels, read as their historical mean; π_5 the live display seen by 30%; π_6 an exponential moving average forecast; π_7 the CE recommendation (Prop. 1); the optimum as advice; and π_8 the optimum plus the reward of Prop. 3. Recommendation users are rational and disobey whenever it saves more than 6 s. Behaviour: logit response with habit, where a share *revise* reconsiders each day. The three regimes are *calm* (λ=3, revise 0.3), *reactive* (λ=3, revise 1) and *noisy* (λ=1, revise 0.3). Learners: Hedge and regret matching (the app publishes every slot's wait) and EXP3 (own wait only). Metrics: mean cost, 90th-percentile wait, *swing* (share of the crowd whose slot load moved versus the previous day), and external and swap regret.

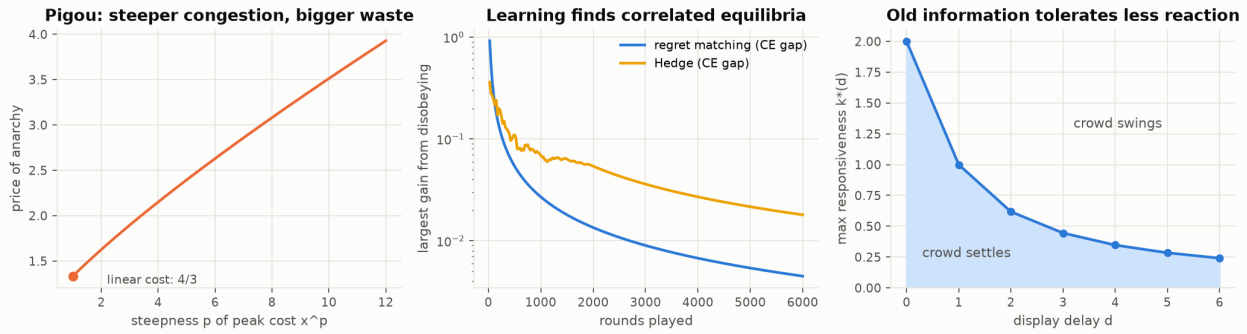


Figure 1: Theory. Left: PoA of Pigou's network grows with steepness. Middle: in the intersection game, regret matching drives the CE gap of the empirical play to zero; Hedge reaches the coarse CE set. Right: stability limit $k^*(d)$ for $y_{t+1} = y_t - k\, y_{t-d}$ (matches the numeric root analysis to 10^{-6}).

4.4 Case study and synthetic data

Four messes (760/750/745/745 students), breakfast 07:30, lunch 12:00 and dinner 19:30, each 12 ten-minute slots. Attendance is 60/85/90%, varying by student and with $\pm 10\%$ whole-day shocks. Preferences: a lunch rush about 13:05 after classes, a dinner rush about 20:10, plus smaller groups. $\beta \in \{1.5, 3, 6\}$ min/slot. Serving rate: 8–9 students/min. On special-menu days (Sunday lunch and breakfast, Wednesday and Sunday dinner) attendance is 5 points higher, the counter runs at 80% speed, and preferences move 10 minutes earlier. We simulate the status quo (no display, logit students with memory) for 8 weeks and write a QR-scan log (`date, meal, mess, menu, student_id, scan_time`), where scan time = arrival + exact queue wait (A2): **393,747 rows**. A hidden ground-truth table scores our estimators. **Calibration:** while a queue exists, plates leave at rate μ, so the busiest 5-minute scan rate estimates μ; a student's median scan slot on normal days estimates τ_i. **Nowcasting:** predict the scans in the next three slots from tonight's scans so far, with N0 (historical mean profile) or N1 (N0 scaled by tonight-so-far over history-so-far), using rolling-origin validation.

5 Results

RQ1: what selfish timing costs. Table 2 and Fig. 2. A normal dinner loses 1.81 min per student (PoA 6.7) and a special dinner 7.57 min (PoA 4.6). Across the four messes this is **132 student-hours per normal day**. Normal breakfasts have no queue at all. The planner removes waiting entirely by feeding arrivals at exactly the serving rate, at the price of moving some students one or two slots. The bumpy Nash profile is genuine: the slot right after a rush inherits its backlog, so arriving at the start of a rush is better than just after it. The same 3–8× gap appears in every domain, and Pigou's 4/3 is recovered exactly when the slot model uses linear costs (unit test).

Table 2: Selfish (Nash) vs. planner, minutes per person (mess rows: average of 4 messes). Last three columns (Prop. 3): share of people tempted to disobey the optimum, average obedience reward per person, and time saved per person vs. Nash.

Instance	Nash	Optimum	PoA	Tempted	Reward	Saving
Mess lunch, normal / special	1.39 / 4.87	0.21 / 1.25	6.7 / 3.9			
Mess dinner, normal / special	2.14 / 9.71	0.33 / 2.14	6.7 / 4.6			
Road: bridge commute (2400 cars)	17.12	4.50	3.8	52%	4.48	12.62
Network: night backups (600 jobs)	3.71	1.13	3.3	56%	1.09	2.58
Restaurant: food court (450)	2.35	0.50	4.7	20%	0.48	1.85
Gym: evening (240)	1.68	0.31	5.4	13%	0.27	1.37
Mess dinner, generic (680)	2.21	0.28	8.0	11%	0.27	1.93

RQ2: what the display should show. Fig. 3. *More information can be worse.* For a reactive crowd the exact live counter raises the cost from 2.24 (Nash) to **11.55 min**, with 76% of the crowd switching slots each day: the rush jumps around, as in the two-student game. Coarse levels do not help (12.95), and private noise helps partly (6.94 at sd 3 and 3.2 at sd 8, Fig. 4). Showing the counter to at most half the crowd costs almost nothing (2.11–2.16), whereas showing it to 70% raises the cost to 4.95: **H1 holds for a reactive crowd**. For calm crowds no information is cheapest (1.65), but only because people keep pessimistic memories. Their positive regret (0.37 min) means the outcome lasts only

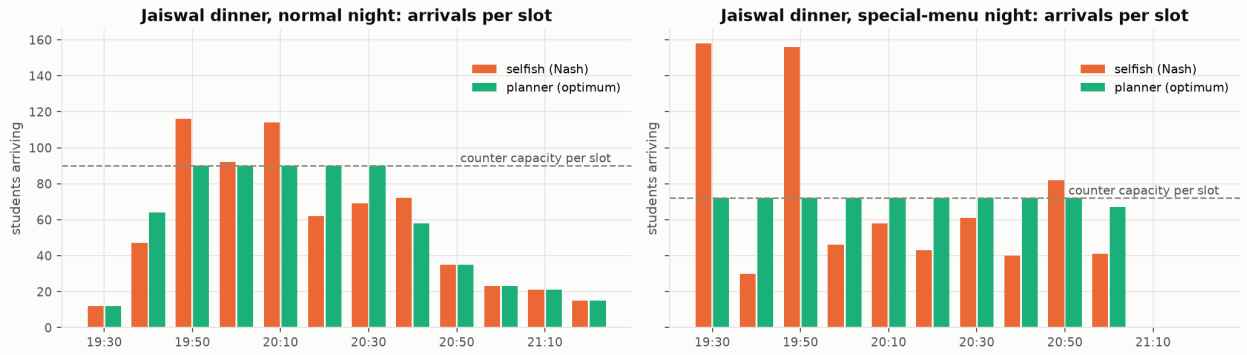


Figure 2: Arrivals per slot at Jaiswal dinner: selfish (orange) overloads the counter (dashed); the planner (green) smooths arrivals to capacity.

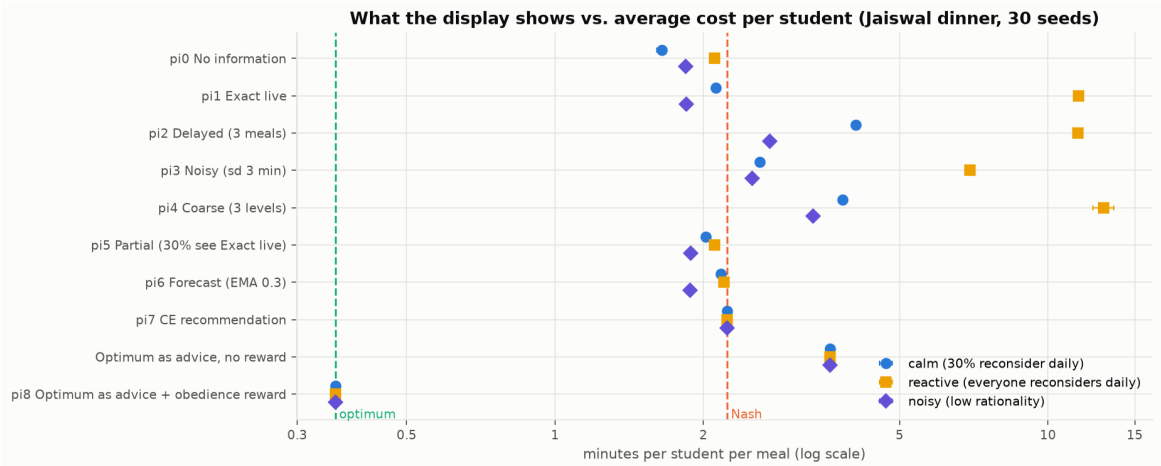


Figure 3: Average cost under each policy and behaviour regime (Jaiswal dinner, 30 seeds, 95% CI; log scale). The CE recommendation (π_7) gives the same result in every regime; only π_8 reaches the optimum.

while they never learn. **H2 (delay).** With 30% of people reacting daily the swing grows from 2% (delay 1) to 11% (delay 4) (Fig. 4), as predicted by $k^*(d)$ shrinking with d.

RQ3: correlated recommendations. Fig. 5. The Nash-based recommendation is a CE: nobody gains by disobeying (0.0 s at Jaiswal dinner). Its outcome is the same in all three regimes, with zero swing and zero regret. With the live display as the fallback for non-users, it removes the herding of a reactive crowd once 50% (2.53) to 70% (2.12) of students use it. The optimum, by contrast, is *not* a CE: 13% of students would gain up to 2.9 min by disobeying, and when rational students do disobey the cost *rises* to 3.61. The best-CE LP over 26 schedules returns exactly the Nash cost (2.235), and so does the best CCE, confirming Prop. 2: advice coordinates but cannot make people accept worse slots for others. Prop. 3 closes the gap. Rewarding the 13% sent off-peak with about 2.7 minutes' worth of priority (0.35 min per student on average) makes the optimum self-enforcing (π_8: **0.36 min**, a 6× reduction). **H3 is refined**: recommendations close the NE–optimum gap only together with obedience rewards.

Learning, incomplete information, data. With published waits, Hedge and regret matching settle near Nash (2.29 / 2.31 min). Their swap regret falls to 0.19 / 0.03 min per day, i.e. joint play approaches a CE, as theory predicts. EXP3, which only sees its own wait, still has 5.1 min of regret after 300 days: publishing data helps learning (Fig. 6). In the Bayesian analysis (slow night with probability 0.3, counter at 70%), announcing the night lowers the expected cost from 4.47 to 3.37 (+1.10 min per student). On *normal* nights, however, it raises the cost from 0.88 to 2.24, because the fear of a slow night was keeping people spread out. Calibration from scans alone recovers μ within 2–3% whenever queues form (it is only a lower bound when they never do), and recovers preferred slots for 97% of students within one slot at dinner (Table 3).

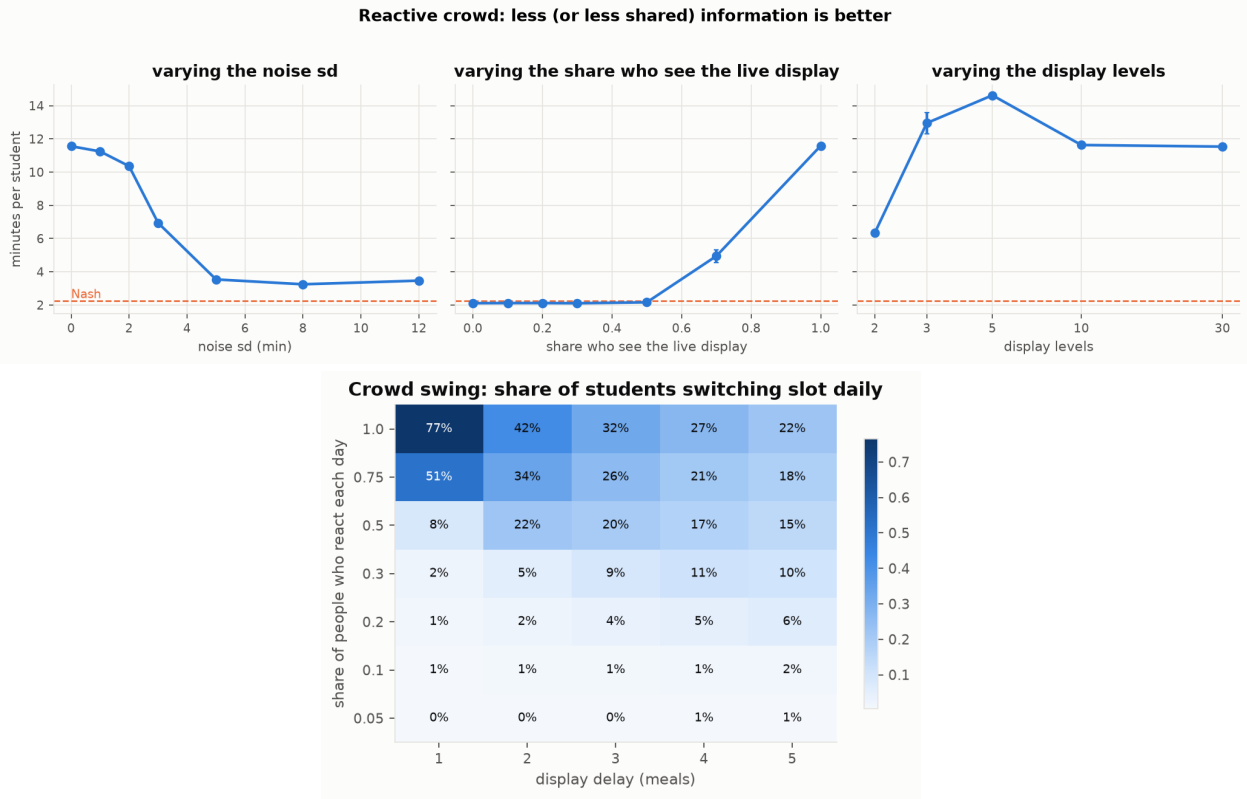


Figure 4: Top: reactive crowd, cost vs. private noise, share of viewers and number of display levels (dashed: Nash). Bottom: crowd swing vs. display delay and share of people reacting daily.

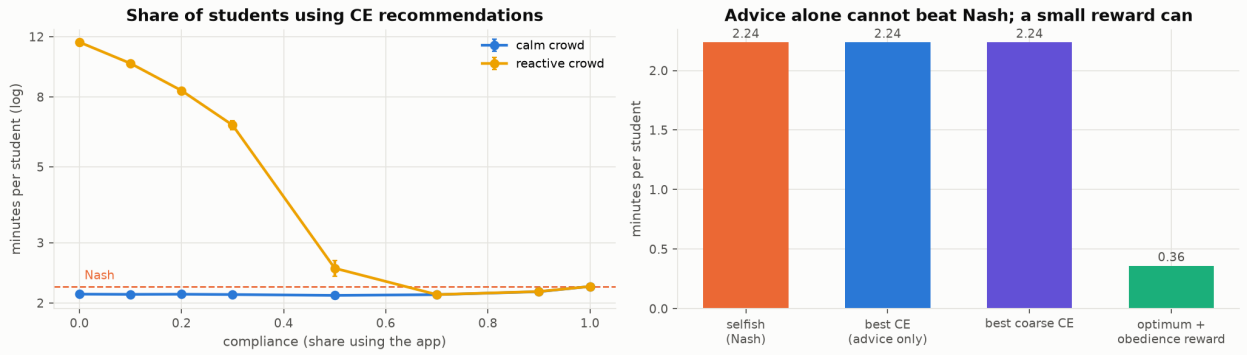


Figure 5: Left: share of students using CE recommendations (non-users see the live display). Right: advice alone (best CE / CCE from the LP) equals Nash; the optimum with the obedience reward is 6× cheaper.

Table 3: Learning unknowns from the scan logs (Jaiswal; nowcast MAE averaged over 4 messes, scans per 10-min slot).

Meal, menu	true μ	estimated μ	mean scans/slot	N0 MAE	N1 MAE
Breakfast, normal	8.0	6.2 (no queue: lower bound)	28.2	3.89	**3.38**
Breakfast, special	6.4	6.6	30.1	**3.64**	3.74
Lunch, normal	9.0	9.2	39.8	**4.02**	4.05
Lunch, special	7.2	7.4	43.7	**3.36**	3.56
Dinner, normal	9.0	9.2	42.3	**4.41**	4.58
Dinner, special	7.2	7.4	43.4	4.37	**3.63**

RQ4: nowcasting. Both forecasters predict the next 30 minutes within 8–14% of the mean load. Scaling by tonight's scans (N1) helps most when an evening starts unusually (normal breakfasts -13%, special dinners -17%); elsewhere

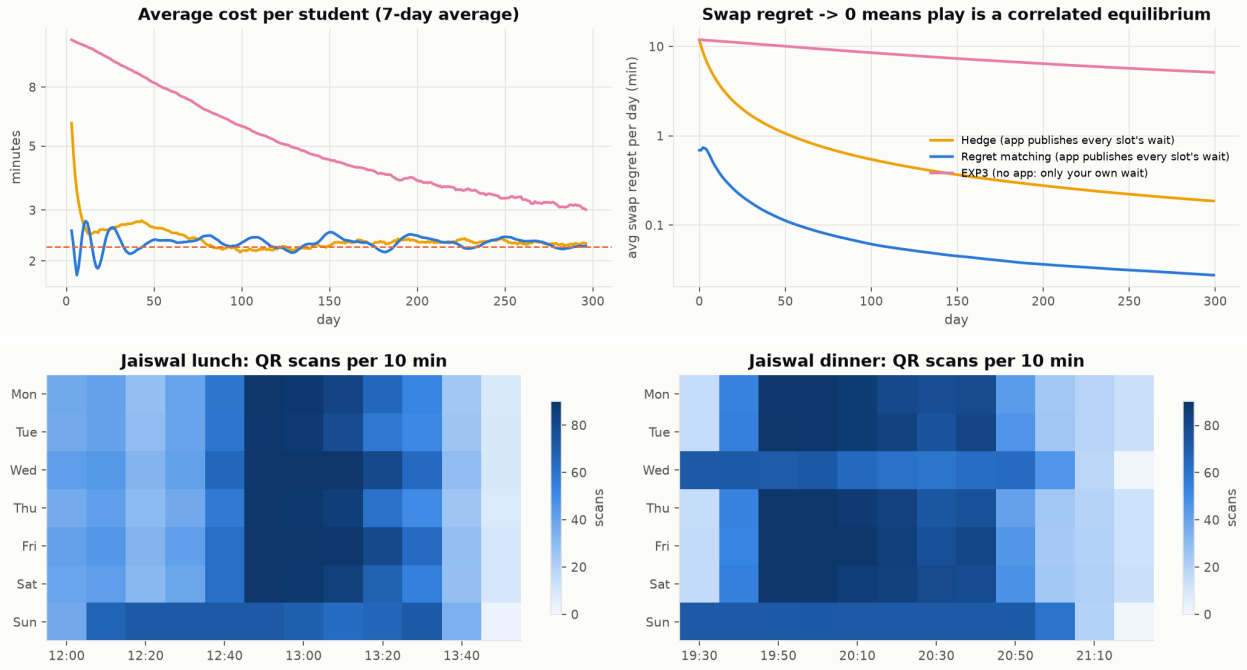


Figure 6: Top: learning over 300 days (dashed: Nash). Bottom: synthetic QR scans per 10 minutes by weekday; the plateaus are the serving rate (Sunday lunch, Wednesday and Sunday dinner are special).

the two are within 5%, as the extra information in a few early scans is offset by their noise. Preferred slots are recovered exactly for 80% of students at dinner (97% within one slot).

6 Limitations and future work

All data are synthetic, so the conclusions are about mechanisms (herding, obedience, the price of obedience), not exact IITGN minutes. The fluid queue spreads arrivals evenly, so the optimum's zero wait (and hence the PoA) is optimistic. With carry-over the game has no potential, so we report ε-equilibria (at most 14 s). The optional scarcity penalty makes the game lumpier (ε up to about 2 min), so it is off by default. Behaviour models are assumptions, and we report which conclusions change across them. Future work: a Telegram bot pilot in one mess; dosa tokens or pre-booked slots as non-monetary tolls; choice between messes (a routing game, where Braess effects are possible); differentially private displays; and a second scan at the queue entrance to measure waits.

7 Contribution

The work was split into four packages, one per member, following the plan in our project brief. All four members took part in the modelling decisions, the analysis of results and the writing of this report.

Table 4: Work packages, what each member delivered (files in `crowd/` and experiments), and status.

Member	Work package and deliverables	Status
Harsh Kumar Jha	**Data and case study.** Mess scenario (4 messes, 3 meals, special menus), synthetic roster and QR-scan logs, serving-rate and preference calibration, nowcasting N0/N1 (`mess.py`; E1, E9).	**Fulfilled**[a]
Yash Goyal	**Incomplete information, learning and statistics.** Bayesian game and value of information; Hedge, EXP3 and regret-matching learners; 30-seed confidence intervals; Streamlit demo, README and repository (`correlated.py`, `simulate.py`, `app.py`; E7, E8).	**Fulfilled**
Piyush Kumar Keshri	**Equilibria and correlated equilibrium.** Toy games; NE, CE and CCE linear programs; Nash, Wardrop and optimum solvers; price of anarchy; CE recommender, Propositions 1–3, best-CE LP and obedience reward (`theory.py`, `game.py`, `correlated.py`; E0, E2, E6).	**Fulfilled** (extended with π_8)
Garv Singhal	**Simulator and information policies.** Day-to-day simulator, policies π_0–π_8, behaviour regimes, sweeps, delay and crowd-swing analysis, generic domains (`simulate.py`, `domains.py`; E3–E5).	**Fulfilled**

[a] As agreed with our advisor, real scan logs were replaced by a synthetic semester, and the manual queue-timing study by the generator's ground-truth waits.

Reproducibility: `pip install -r requirements.txt; pytest -q` (54 checks); `python experiments.py` regenerates every figure and number in this report (`results/RESULTS.md`); `streamlit run app.py` opens the demo, which is also hosted at https://crowd-congestion.streamlit.app/.

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